

Twelve-critical graphs with $\left(\frac{2}{5} + o(1)\right) n^2$ edges

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Abstract

Let $f_k(n)$ denote the maximum number of edges in a k -critical graph on n vertices. We construct a sequence of twelve-critical graphs G_s with $|V(G_s)| \rightarrow \infty$ and

$$e(G_s) = \left(\frac{2}{5} + o(1)\right) |V(G_s)|^2.$$

Since $2/5 > 3/8$, this disproves $f_{12}(n) \sim 3n^2/8$, and hence the general asymptotic formula proposed in Erdős Problem 917. The construction combines a coloring module of Pegden with K_5 -saturated graphs of sublinear maximum degree.

1 Introduction

All graphs are finite and simple. A graph is k -critical if it has chromatic number k and every proper subgraph is $(k-1)$ -colorable; for graphs without isolated vertices this is equivalent to requiring only that deleting any edge lowers the chromatic number, the convention of [4]. Write $e(G) = |E(G)|$, and let $f_k(n)$ be the largest edge count on n vertices under that convention. Erdős [3, p. 28] conjectured that, for fixed $k \geq 6$ (in the formulation of [4]),

$$f_k(n) \sim \frac{1}{2} \left(1 - \frac{1}{\lfloor k/3 \rfloor}\right) n^2. \quad (1.1)$$

Dirac [2] gave the $k = 6$ construction obtained by joining two odd cycles of equal order, and Erdős observed that it generalizes when $3 \mid k$, giving the coefficient in (1.1). When $3 \nmid k$, Toft's constructions [8] give larger coefficients. For background on dense critical graphs, see Jensen and Toft [6, Section 5.1] and Jensen [5]. Luo, Ma, and Yang [7] noted in 2023 that no constructions improving Toft's asymptotic constants had been found since 1970. We show that the coefficient in (1.1) is not optimal for $k = 12$.

Theorem 1. *There is a sequence of twelve-critical graphs G_s such that*

$$|V(G_s)| \rightarrow \infty, \quad \frac{e(G_s)}{|V(G_s)|^2} \rightarrow \frac{2}{5}.$$

In particular, $f_{12}(n) \not\sim 3n^2/8$.

The graph G consists of five copies of Pegden's module $U(S, T)$ [9], with their active sets joined by an almost complete five-partite graph. The omitted edges form a graph Z whose adjacency is prescribed by a K_5 -saturated graph H of small maximum degree from [1]. In an eleven-coloring each active set needs three colors (Lemma 2); the degree bound forces two of them to be private to that part. The eleventh color must then appear in all five parts, giving a K_5 in Z , impossible since H is K_5 -free. Deleting an edge between two parts adds an edge to Z , and saturation supplies a K_5 through it, allowing one color to be shared in an eleven-coloring of $G - e$. Complete joins of critical graphs, as in Dirac's construction, cannot share colors between factors; omitting Z costs $o(n^2)$ edges and permits this shared color. Deleting the $o(n^2)$ module edges leaves a five-partite graph, so the asymptotic density of this construction is at most $2/5$. For comparison, the upper bound of Luo, Ma, and Yang [7, Theorem 1.1] has coefficient $9/20 - 1/4356 \approx 0.44977$ for f_{12} .

2 A coloring module

Let S be 11-critical and T be 10-critical. The module $U(S, T)$ consists of disjoint copies of S and T , together with an independent set

$$A = V(S) \times V(T).$$

Each $(x, y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T , and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x .

The following is the $U(11, 10)$ case of Pegden's module lemma [9, Lemma 2.5], stated there for triangle-free factors; the proof below does not use that hypothesis.

Lemma 2. *Fix a palette of eleven colors.*

1. *In every proper coloring of $U(S, T)$ from this palette, the active set uses at least three colors.*
2. *Given $a_0 = (x_0, y_0) \in A$ and distinct colors α, β, γ , there is a proper coloring in which the active set uses only α, β, γ , and γ occurs on the active set only at a_0 .*
3. *Given an edge e of $U(S, T)$ and distinct colors α, β , there is a proper coloring of $U(S, T) - e$ in which the active set uses only α, β .*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all eleven colors, it contains a vertex colored α and a vertex colored β . Every active vertex in the first row must have color β , so no vertex of T can have color β . The second row similarly excludes α from T . This leaves only nine colors for T , a contradiction.

For (2), color S so that α occurs only at x_0 . Color T with the ten colors other than α , so that β occurs only at y_0 . Extend these colorings by

$$\text{col}(x, y) = \begin{cases} \alpha, & x \neq x_0, \\ \beta, & x = x_0, y \neq y_0, \\ \gamma, & (x, y) = (x_0, y_0). \end{cases} \quad (2.1)$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e \in E(S)$, color both $S - e$ and T with the ten colors other than α , and color all of A with α . If $e \in E(T)$, color $T - e$ with the nine colors other than α, β . Choose $x_0 \in V(S)$ and color S with α only at x_0 . Color the row indexed by x_0 with β and all other rows with α .

Suppose instead that e joins an active vertex (x_0, y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0, y_0) to x_0 , use (2.1) with the color of (x_0, y_0) changed to α . If it joins (x_0, y_0) to y_0 , change that color to β . In either case the only edge that would become monochromatic is the deleted edge. $\square$

3 From saturated graphs to twelve-critical graphs

A graph is K_5 -saturated if it contains no K_5 , but adding any missing edge creates a K_5 . Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a triangle.

Proposition 3. *Let H be a K_5 -saturated graph on $v \geq 5$ vertices, with maximum degree $d < v - 1$. Let $h \geq 11$ be odd and suppose*

$$v > 40hd. \quad (3.1)$$

Put $a = 2hv$. There is a twelve-critical graph G with

$$|V(G)| = 5(a + h + 2v) \quad (3.2)$$

and

$$e(G) \geq 10a^2 \left(1 - \frac{d}{v}\right). \quad (3.3)$$

Proof. Construction. Set $\ell = 2v$ and take

$$S = K_8 \vee C_{h-8}, \quad T = K_7 \vee C_{\ell-7}.$$

Both cycles are odd and have length at least three. Since the join of a p -critical and a q -critical graph is $(p+q)$ -critical [2], S is 11-critical and T is 10-critical. (Deleting a vertex or an edge within one factor saves a color there; if a joining edge xy is deleted, give x and y singleton colors in their respective factors and identify those colors.)

Take five disjoint copies $U_1, \dots, U_5$ of $U(S, T)$, with active sets $A_1, \dots, A_5$, each of size $a = h\ell$. In each copy, identify $V(T)$ with $V(H) \times \{0, 1\}$ by any bijection. An active vertex then has the form $(x, (z, \varepsilon))$; give it label z . Let π denote this labeling map. Each label occurs exactly $b = 2h$ times in each active set.

Define a five-partite graph Z on $A_1 \cup \dots \cup A_5$ by putting, for $u \in A_i$ and $w \in A_j$ with $i \neq j$,

$$uw \in E(Z) \iff \pi(u)\pi(w) \in E(H).$$

The graph G retains all the edges of the five modules and, between different active sets, takes the complement of Z . There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G .

We record three properties of Z . First, Z is K_5 -free: the labeling map sends every clique injectively to a clique of H . Moreover, adding any missing edge between different parts of Z creates a K_5 with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a triangle in $N_H(z) \cap N_H(z')$. Choose vertices with these three labels in the other three parts. Together with u, w , they form the required clique in $Z + uw$. If the labels coincide, say both equal z , choose a vertex z' not adjacent to z ; this is possible because $d < v - 1$. The same common-neighborhood triangle is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_4 across any prescribed four parts. Indeed, H has a missing edge; one endpoint together with a triangle in the common neighborhood gives a K_4 in H . Its four labels can be placed in any four parts of Z .

Third, $D := \Delta(Z) = 4bd = 8hd$, so (3.1) gives $\ell > 10D$.

The chromatic lower bound. Suppose that G has an eleven-coloring. Fix a part A_i and any color α . Since its copy of S uses all eleven colors, some vertex x of S has color α . The ℓ active vertices in the row indexed by x avoid α , so the pigeonhole principle and $\ell > 10D$ give a color $\beta \neq \alpha$ occurring more than D times in that row. Applying this observation twice, we may choose two distinct colors α_i, β_i , each used more than D times in A_i .

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Thus the ten chosen colors $\alpha_1, \beta_1, \dots, \alpha_5, \beta_5$ are distinct, leaving a single color γ . The active set A_i can use only $\alpha_i, \beta_i, \gamma$, and Lemma 2(1) forces γ to occur in every part. Choosing a vertex of this color in each part gives a K_5 in Z , a contradiction. Thus $\chi(G) \geq 12$.

Colorings after edge deletion. Use an eleven-color palette

$$\alpha_1, \beta_1, \dots, \alpha_5, \beta_5, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G . By the first property of Z , there is a K_5 in $Z + uw$ containing uw , with vertices $a_i \in A_i$. Apply Lemma 2(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i . The only color shared by different active sets is γ . The five vertices of this color are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained an eleven-coloring of $G - e$.

Now let e lie within U_i . Apply Lemma 2(3) to color $U_i - e$, using only α_i, β_i on A_i . Choose a K_4 in Z across the other four parts. In each of those four modules, apply Lemma 2(2) with the selected vertex as the only active vertex of color γ . Once again this is a proper eleven-coloring of $G - e$.

These cases cover every edge of G , so $G - e$ is eleven-colorable for every edge e ; recoloring one endpoint of a deleted edge with a twelfth color gives $\chi(G) \leq 12$. Since G has no isolated vertices, every proper subgraph of G lies in some $G - e$. Hence G is twelve-critical.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each, where $m = e(H)$. Since $2b^2m \leq b^2vd$ and $a = bv$, at least $10a^2(1 - d/v)$ edges lie between active sets, proving (3.3). $\square$

Remark (Exact edge count). The construction in Proposition 3 satisfies

$$e(G) = 10(a^2 - 8h^2m) + 5(2a + 9h + 16v - 79), \quad m = e(H). \quad (3.4)$$

Indeed, each pair of active sets contributes $a^2 - 2b^2m$ edges, while $e(S) = 9h - 44$, $e(T) = 16v - 35$, and each module has $2a$ edges incident with its active set.

4 Choice of the saturated graphs

We use the following instance of the construction in [1, Section 4, Example 2].

Lemma 4 (Alon–Erdős–Holzman–Krivelevich). *For every prime power $Q \geq 4$, there is a K_5 -saturated graph H_Q with*

$$|V(H_Q)| = 13Q^2 + 12Q, \quad \Delta(H_Q) = 22Q - 3. \quad (4.1)$$

Proof. In Example 2 of [1], set $k = 5$ and give each of the Q^2 additional independent sets one vertex. The core has $12Q(Q + 1)$ vertices. A core vertex has $3(3Q - 1)$ neighbors in its level, $12Q$ in the other levels, and Q among the additional vertices, so its degree is $22Q - 3$. An additional vertex has degree $12(Q + 1)$, which is smaller for $Q \geq 4$. Thus (4.1) holds; saturation is part of the cited construction. $\square$

Proof of Theorem 1. For each integer $s \geq 4$, set

$$h = 3^s, \quad Q = h^2,$$

and take $H = H_Q$ from Lemma 4. Write $v = |V(H)|$ and $d = \Delta(H)$. Then

$$v = 13h^4 + 12h^2, \quad d = 22h^2 - 3 < v - 1.$$

The integer h is odd and at least 81. Moreover,

$$40hd < 880h^3 < 13h^4 \leq v,$$

since $13h \geq 13 \cdot 81 > 880$. Proposition 3 therefore gives a twelve-critical graph G_s . Put $a = 2hv$ and $N_s = |V(G_s)|$. We have

$$\frac{N_s}{5a} = 1 + \frac{1}{h} + \frac{1}{2v} \longrightarrow 1, \quad \frac{d}{v} \longrightarrow 0. \quad (4.2)$$

There are at most $10a^2$ edges between active sets, and the five modules contain $5(2a + 9h + 16v - 79) = O(a)$ edges. Together with (3.3), this gives

$$10a^2 \left(1 - \frac{d}{v}\right) \leq e(G_s) \leq 10a^2 + O(a).$$

Dividing by N_s^2 and using (4.2) yields

$$\frac{e(G_s)}{N_s^2} \longrightarrow \frac{10}{25} = \frac{2}{5}.$$

Finally, $N_s \rightarrow \infty$ and $f_{12}(N_s) \geq e(G_s)$. Since $2/5 > 3/8$, this subsequence rules out $f_{12}(n) \sim 3n^2/8$. $\square$

Remark (Other chromatic numbers). For each fixed $t \geq 3$, the same construction uses t modules, with

$$S = K_{2t-2} \vee C_{h-2t+2}, \quad T = K_{2t-3} \vee C_{2v-2t+3},$$

of chromatic numbers $2t + 1$ and $2t$, and a K_t -saturated graph H of order v and maximum degree d . Take $h \geq 2t + 1$ odd and $v \geq t$, with $d < v - 1$ and $v > 2t(t - 1)hd$. Then $D = 2h(t - 1)d$ and $2v > 2tD$. In a $(2t + 1)$ -coloring, each part has two private colors, and the remaining color would give a K_t in Z . Saturation supplies the transversal K_t after a cross-part edge deletion, and a K_{t-1} in the other parts after an internal deletion. The module colorings therefore give a $(2t + 2)$ -critical graph. By [1, Theorem 7], we may take $v \rightarrow \infty$ with $d = O_t(\sqrt{v})$, then choose odd $h \rightarrow \infty$ with $h = o(\sqrt{v})$. Writing $a = 2hv$, the order is $t(a + h + 2v) \sim ta$, the cross-part edge count is $\binom{t}{2}a^2(1 - o(1))$, and the module edges contribute $O_t(a)$. Thus every even $k = 2t + 2 \geq 8$ admits a sequence with density

$$\frac{t - 1}{2t} = \frac{k - 4}{2(k - 2)}.$$

Joining these graphs to K_1 gives, for every odd $k \geq 9$, a k -critical sequence with density $(k - 5)/(2(k - 3))$.

Lean formalization

Lemma 2, Proposition 3 (criticality, order, and the exact edge count (3.4)), the parameter choices, and the limit $e(G_s)/|V(G_s)|^2 \rightarrow 2/5$ have been formalized in Lean 4 with Mathlib. The formalization also defines f_{12} under the edge-deletion convention and proves that $f_{12}(n)/n^2$ has no limit below $2/5$. The only unformalized input to these results is the existence of the graphs H_Q of Lemma 4, stated as an explicit hypothesis. Source and build instructions: <https://github.com/FireflySentinel/erdos-917>.

Use of generative AI

The proofs and the first draft were generated with GPT-6 Astra; GPT-5.6 Sol and Claude Opus 5 were used for editorial review; the Lean formalization was developed with OpenAI Codex (GPT-6). The author checked the arguments and is responsible for the content.

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